

YEAR 7 MATHEMATICS

Chapter 6

Geometrical figures

MIXED PRACTICE WORKSHEETS

6

How to use this booklet

- Each lesson is **one double-sided sheet**. Write your name, class and date at the top.
- Then work down the **columns**, left to right. They get harder as you go: **Easy** **Easy** **Medium** on page 1, then **Medium** **Challenging** **Challenging** on page 2.
- Most questions practise **today's skill**. A question marked **from 1.03** comes from an earlier lesson, to keep it fresh. Stuck? Look back at that lesson.
- Each question tells you what to do: read it carefully. Show your working on the lines. The first question with working in each Easy column shows you how to set it out.

WE DO

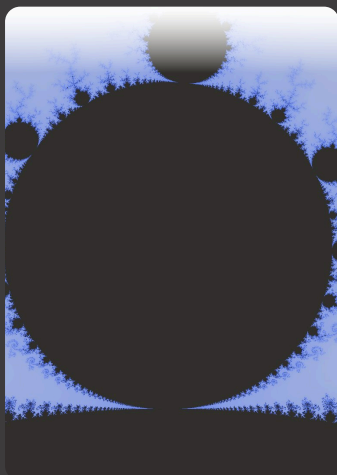
The grey box at the top. Your teacher shows the key idea with two or three examples. **Copy every step** into the space.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
6.01	Transforming shapes	3		☐
6.02	Combined transformations	5		☐
6.03	Line symmetry	7		☐
6.04	Rotational symmetry	9		☐
6.05	Classifying triangles	11		☐
6.06	Angle sum of a triangle	13		☐
6.07	Exterior angle of a triangle	15		☐
6.08	Quadrilaterals	17		☐
6.09	Angle sum of a quadrilateral	19		☐
6.10	Properties of quadrilaterals	21		☐

In this chapter you will

- reflect, translate and rotate shapes
- find line and rotational symmetry
- classify triangles and quadrilaterals
- use the angle sums of triangles and quadrilaterals
- use the exterior angle of a triangle
- describe the properties of special quadrilaterals



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is the top bulb of the set and the lacy edge around it. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

NAME _____ CLASS _____ DATE _____

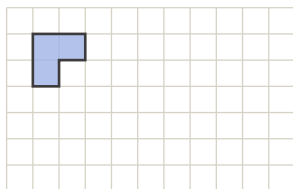
WE DO

KEY IDEA

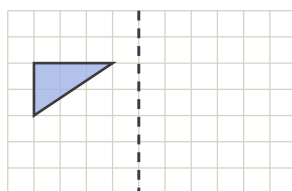
A translation slides a shape. A reflection flips it over a mirror line. A rotation turns it about a point. The shape keeps its size and shape: only its position changes.

Watch your teacher.
Copy every step.

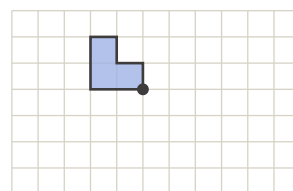
- E1** Translate the shape 4 right and 2 down. Draw the image.



- E2** Reflect the shape in the dashed line.



- E3** Rotate the shape 90° clockwise about the dot.

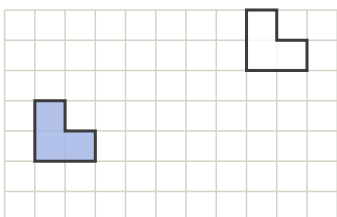


Easy

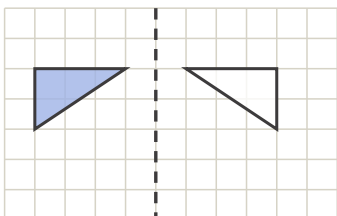
Questions 1–4

- 1** Is a slide a translation, a reflection or a rotation?

- 2** Describe the translation from the shaded shape to the white shape.



- 3** Name the transformation from the shaded shape to the white shape.

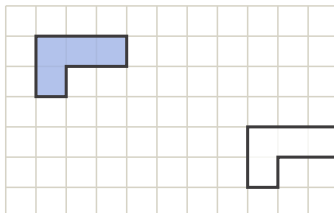


- 4** Is a flip a translation, a reflection or a rotation?

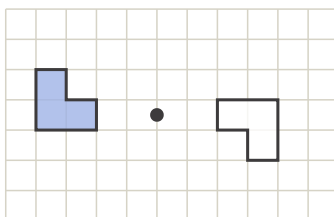
Easy

Questions 5–8

- 5** Describe the translation from the shaded shape to the white shape.



- 6** Name the transformation from the shaded shape to the white shape.



- 7** Is a mirror image a translation, a reflection or a rotation?

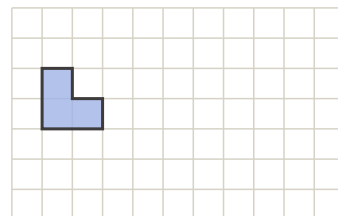
- 8** Use a calculator to write $\frac{3}{16}$ as a decimal.

from 4.09

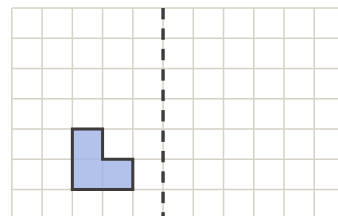
Medium

Questions 9–13

- 9** Translate the shape 4 right and 2 down. Draw the image.



- 10** Reflect the shape in the dashed line. Draw the image.

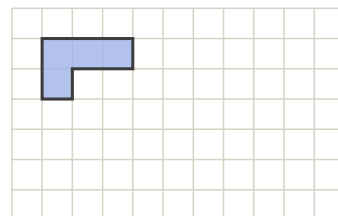


- 11** Solve in your head: $\frac{k}{6} = 12$

from 5.10

- 12** A point is 3 squares to the left of a mirror line. Where is its image?

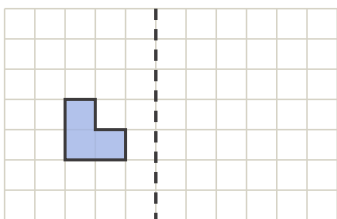
- 13** Translate the shape 4 right and 2 down. Draw the image.



Medium

Questions 14–21

- 14 Reflect the shape in the dashed line. Draw the image.

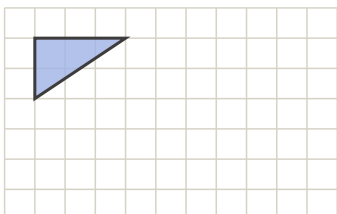


- 15 A point is 6 squares to the left of a mirror line. Where is its image?

- 16 Which is larger: $\frac{1}{10}$ or $\frac{1}{8}$?

from 4.02

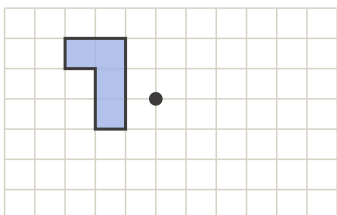
- 17 Translate the shape 6 right. Draw the image.



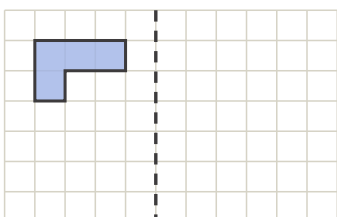
- 18 $P = 2l + 2w$. Find P when $l = 14$ and $w = 3$.

from 5.05

- 19 Rotate the shape 90° clockwise about the dot. Draw the image.



- 20 Reflect the shape in the dashed line. Draw the image.



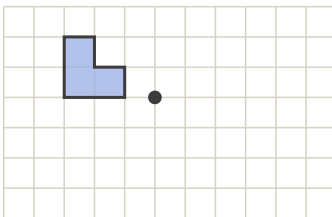
- 21 A number plus 5 is 22. What is the number?

from 5.13

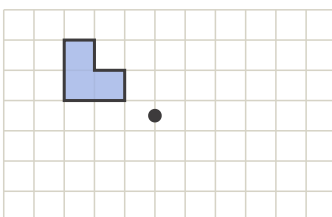
Challenging

Questions 22–28

- 22 Rotate the shape 90° clockwise about the dot. Draw the image.



- 23 Rotate the shape 180° about the dot. Draw the image.



- 24 $bx \div b$

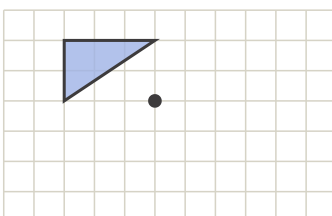
from 5.09

- 25 A shape is turned 90° clockwise, then another 90° clockwise. What single turn does the same?

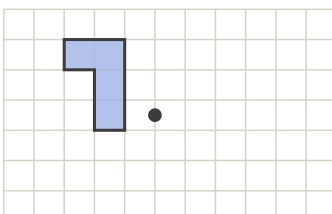
- 26 Solve $\frac{m}{2} + 6 = 13$.

from 5.12

- 27 Rotate the shape 90° clockwise about the dot. Draw the image.



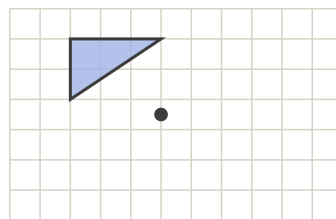
- 28 Rotate the shape 180° about the dot. Draw the image.



Challenging

Questions 29–36

- 29 Rotate the shape 180° about the dot. Draw the image.

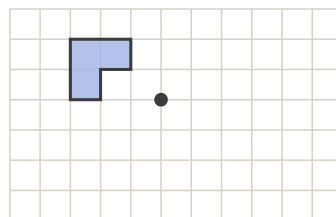


- 30 A shape is turned 90° clockwise. What turn brings it back?

- 31 How many $\frac{1}{5}$ s are in 4? Work out $4 \div \frac{1}{5}$.

from 4.07

- 32 Rotate the shape 90° clockwise about the dot. Draw the image.

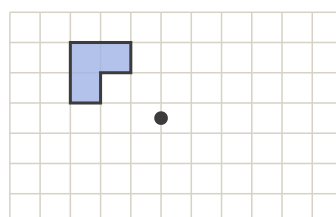


- 33 $16 + 19 + 84 + 45$

from 5.01

- 34 A point is 2 squares to the left of a mirror line. Where is its image?

- 35 Rotate the shape 180° about the dot. Draw the image.



- 36 $5\frac{7}{8} - 3\frac{3}{8}$

from 4.04

NAME _____ CLASS _____ DATE _____

WE DO

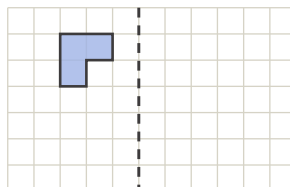
Watch your teacher.
Copy every step.

KEY IDEA

In a combined transformation, do the moves one at a time, in order. Two translations can be done as one: add up the moves across, and the moves up and down.

- E1** Translate 3 right, 1 up, then 2 right, 4 down. Write it as one translation.

- E2** Reflect the shape in the dashed line, then translate the image 2 down. Draw both.



- E3** A shape is turned 90° clockwise, then 180° . What single turn is that?

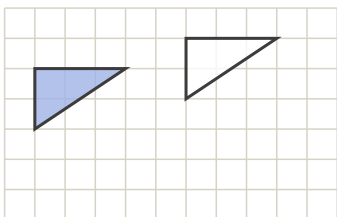
Easy

Questions 1–5

- 1** Translate 3 right, then 2 right. Write it as one translation.

- 2** A shape is turned 90° clockwise, then 270° clockwise. What single turn does the same?

- 3** Name the transformation from the shaded shape to the white shape.



- 4** Translate 2 right, then 7 right. Write it as one translation.

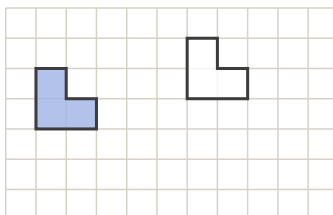
- 5** A shape is turned 90° clockwise, then 180° clockwise. What single turn does the same?

Easy

Questions 6–11

- 6** A shape is turned 180° clockwise, then 180° clockwise. What single turn does the same?

- 7** Name the transformation from the shaded shape to the white shape.



- 8** Translate 5 right, then 2 right. Write it as one translation.

- 9** Simplify $5t - 3t$.

from 5.07

- 10** Translate 7 right, 7 up, then 4 left, 5 down. Write it as one translation.

across: ____ - ____ = ____

up/down: ____ - ____ = ____

____ right, ____ up

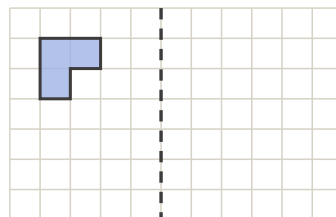
- 11** A shape is turned 90° clockwise, then 90° clockwise. What single turn does the same?

Medium

Questions 12–14

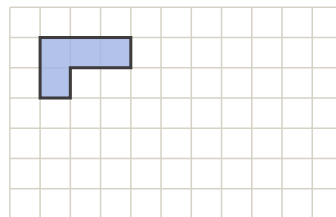
- 12** Translate 6 right, 3 up, then 7 left, 7 down. Write it as one translation.

- 13** Reflect the shape in the dashed line, then translate the image 2 down. Draw both images.



- 14** Translate the shape 3 right and 1 up. Draw the image.

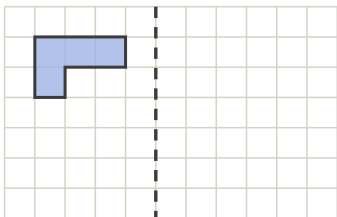
from 6.01



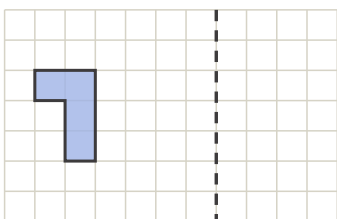
Medium

Questions 15–20

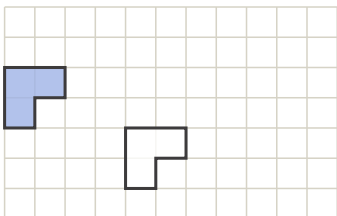
- 15 Reflect the shape in the dashed line, then translate the image 2 down. Draw both images.



- 16 Translate the shape 4 right, then reflect the image in the dashed line. Draw both images.



- 17 Describe the translation from the shaded shape to the white shape.



from 6.01

- 18 Translate 3 right, 3 up, then 5 left, 5 down. Write it as one translation.

- 19 A point is 4 squares to the left of a mirror line. Where is its image?

from 6.01

- 20 Which single transformation is the same as two reflections in the same line?

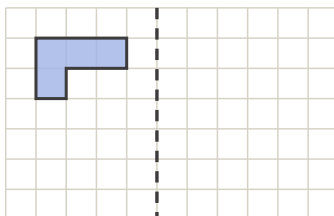
Challenging

Questions 21–27

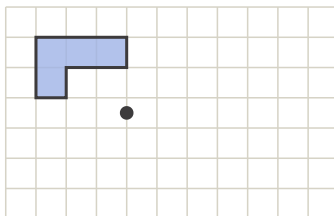
- 21 Does the order matter if you translate 3 right, then reflect in a vertical line? Try it.

- 22 A shape is translated 4 right and 3 up. What translation brings it back?

- 23 Reflect the shape in the dashed line. Draw the image. from 6.01



- 24 Rotate the shape 180° about the dot, then translate the image 3 right. Draw both images.



- 25 $5\frac{3}{4} - 4\frac{1}{4}$

from 4.04

- 26 Does the order matter if you reflect in a line and then translate along it? Try it.

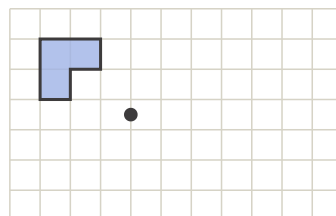
- 27 A shape is translated 3 right and 6 up. What translation brings it back?

Challenging

Questions 28–33

- 28 A shape is translated 5 right and 6 up. What translation brings it back?

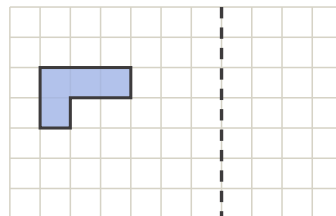
- 29 Rotate the shape 180° about the dot, then translate the image 3 right. Draw both images.



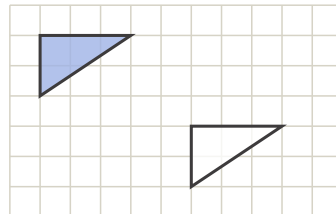
- 30 9 equal packs hold 216 pencils. How many pencils are in each pack?

from 5.13

- 31 Translate the shape 4 right, then reflect the image in the dashed line. Draw both images.



- 32 Describe the translation from the shaded shape to the white shape.



from 6.01

- 33 A shape is translated 3 right and 5 up. What translation brings it back?

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

An axis (line) of symmetry folds a shape onto itself: the two halves match exactly. A regular polygon has as many axes of symmetry as it has sides.

Watch your teacher.
Copy every step.

- E1** How many axes of symmetry does a square have?



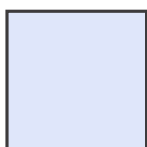
- E2** How many axes of symmetry does the letter E have?

- E3** How many axes of symmetry does a regular hexagon have?

Easy

Questions 1–6

- 1** How many axes of symmetry does the shape have?



- 2** How many axes of symmetry does the capital letter D have?

- 3** How many axes of symmetry does a regular polygon with 7 sides have?

- 4** How many axes of symmetry does the shape have?



- 5** How many axes of symmetry does the capital letter M have?

- 6** How many axes of symmetry does a regular polygon with 6 sides have?

Easy

Questions 7–12

- 7** How many axes of symmetry does the capital letter P have?

- 8** How many axes of symmetry does a regular polygon with 5 sides have?

- 9** How many axes of symmetry does the shape have?



- 10** $cy \div c$

from 5.09

- 11** Name the shape, then draw its axes of symmetry. How many are there?



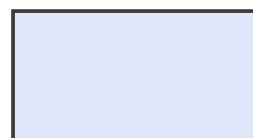
____ axes

- 12** How many axes of symmetry does the capital letter C have?

Medium

Questions 13–17

- 13** Name the shape, then draw its axes of symmetry. How many are there?



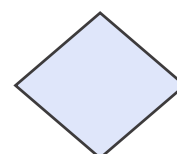
- 14** How many axes of symmetry does the word MUM have (in capitals)?

- 15** A point is 6 squares to the left of a mirror line. Where is its image?

from 6.01

- 16** How many axes of symmetry does a rhombus have?

- 17** Name the shape, then draw its axes of symmetry. How many are there?



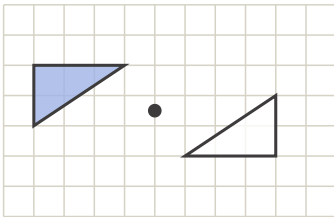
Medium

Questions 18–23

- 18 How many axes of symmetry does the word WOW have (in capitals)?

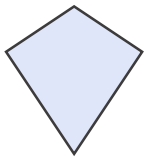
- 19 How many axes of symmetry does a kite have?

- 20 Name the transformation from the shaded shape to the white shape.



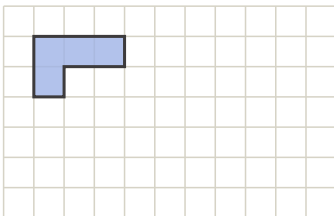
from 6.02

- 21 Name the shape, then draw its axes of symmetry. How many are there?

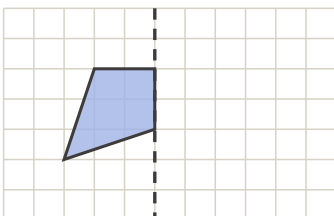


- 22 Translate the shape 5 right and 1 up. Draw the image.

from 6.01



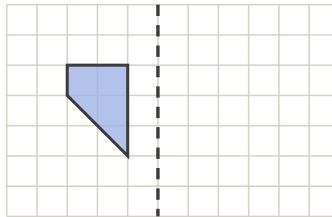
- 23 The dashed line is an axis of symmetry. Draw the other half of the shape.



Challenging

Questions 24–30

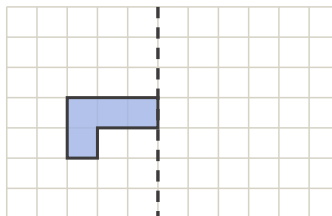
- 24 The dashed line is an axis of symmetry. Draw the other half of the shape.



- 25 How many axes of symmetry does a circle have?

- 26 Reflect the shape in the dashed line. Draw the image.

from 6.01

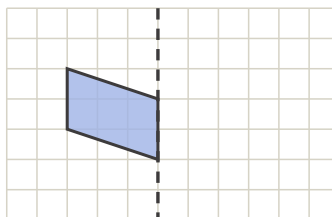


- 27 How many axes of symmetry does a regular octagon have? How many go through corners?

- 28 How many $\frac{1}{4}$ s are in 5? Work out $5 \div \frac{1}{4}$.

from 4.07

- 29 The dashed line is an axis of symmetry. Draw the other half of the shape.



- 30 Can a triangle have exactly two axes of symmetry? Explain.

Challenging

Questions 31–38

- 31 Draw a quadrilateral with exactly one axis of symmetry. Name it.

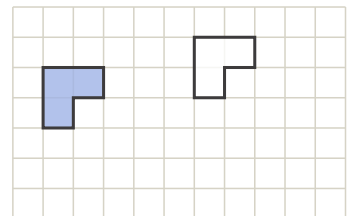
- 32 How many axes of symmetry does a regular 20-sided polygon have? How many go through corners?

- 33 Write an expression: 9 more than 9 times a

from 5.04

- 34 How many axes of symmetry does a rectangle have?

- 35 Name the transformation from the shaded shape to the white shape.



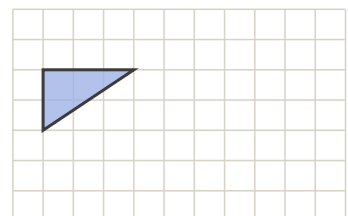
from 6.02

- 36 How many axes of symmetry does a scalene triangle have?

- 37 How many axes of symmetry does an isosceles triangle have?

- 38 Translate the shape 5 right and 2 down. Draw the image.

from 6.01



NAME _____ CLASS _____ DATE _____

WE DO

Watch your teacher.
Copy every step.

KEY IDEA

A shape has rotational symmetry if it fits onto itself before a full turn. The order is how many times it fits onto itself in one full turn. The smallest angle of turn is $360^\circ \div$ the order.

- E1** What is the order of rotational symmetry of a square?



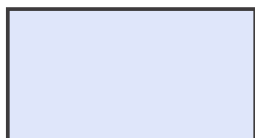
- E2** A regular pentagon: find the order and the smallest angle of turn.

- E3** What is the order of rotational symmetry of the letter S?

Easy

Questions 1–6

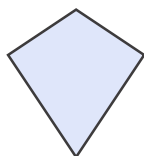
- 1** What is the order of rotational symmetry? (Write 1 if it only fits after a full turn.)



- 2** What is the order of rotational symmetry of the capital letter U?

- 3** What is the order of rotational symmetry of a regular polygon with 10 sides?

- 4** What is the order of rotational symmetry? (Write 1 if it only fits after a full turn.)



- 5** What is the order of rotational symmetry of the capital letter S?

- 6** What is the order of rotational symmetry of a regular polygon with 9 sides?

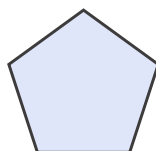
Easy

Questions 7–13

- 7** What is the order of rotational symmetry of the capital letter M?

- 8** What is the order of rotational symmetry of a regular polygon with 8 sides?

- 9** What is the order of rotational symmetry? (Write 1 if it only fits after a full turn.)



- 10** Write 67% as a decimal.

from 4.10

- 11** A regular polygon has 4 sides. Find its order of rotational symmetry and its smallest angle of turn.

order ____

____ $^\circ \div$ ____ = ____ $^\circ$

- 12** What is the order of rotational symmetry of the capital letter F?

- 13** What is the order of rotational symmetry of a regular polygon with 12 sides?

Medium

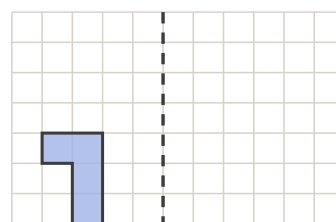
Questions 14–17

- 14** A regular polygon has 24 sides. Find its order of rotational symmetry and its smallest angle of turn.

- 15** A shape has rotational symmetry of order 2. What is its smallest angle of turn?

- 16** Reflect the shape in the dashed line. Draw the image.

from 6.01



- 17** Give the number of axes of symmetry and the order of rotational symmetry of the shape.

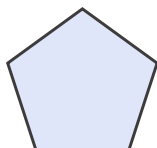


Medium

Questions 18–24

- 18 A shape has rotational symmetry of order 12. What is its smallest angle of turn?

- 19 Give the number of axes of symmetry and the order of rotational symmetry of the shape.



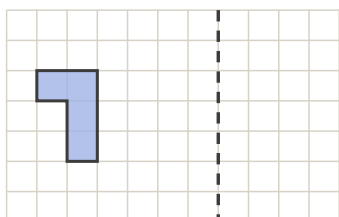
- 20 How many axes of symmetry does a regular polygon with 7 sides have?

from 6.03

- 21 A regular polygon has 15 sides. Find its order of rotational symmetry and its smallest angle of turn.

- 22 Translate the shape 3 right, then reflect the image in the dashed line. Draw both images.

from 6.02



- 23 A shape fits onto itself every 40° . What is its order of rotational symmetry?

- 24 A shape has rotational symmetry of order 10. What is its smallest angle of turn?

Challenging

Questions 25–33

- 25 A shape fits onto itself every 20° . What is its order of rotational symmetry?

- 26 Name a triangle with rotational symmetry of order 3.

- 27 Name the shape, then draw its axes of symmetry. How many are there?



from 6.03

- 28 Compare the line symmetry and the rotational symmetry of a rectangle.

- 29 Simplify $c \times c$.

from 5.03

- 30 A shape fits onto itself every 60° . What is its order of rotational symmetry?

- 31 Name a shape with rotational symmetry of order 4.

- 32 Translate 5 right, then 6 right. Write it as one translation.

from 6.02

- 33 Compare the line symmetry and the rotational symmetry of a kite.

Challenging

Questions 34–40

- 34 Name a capital letter with rotational symmetry of order 2 but no axes of symmetry.

- 35 Compare the line symmetry and the rotational symmetry of a regular hexagon.

- 36 Solve $\frac{f}{5} + 4 = 9$.

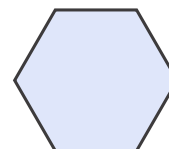
from 5.12

- 37 A shape fits onto itself every 45° . What is its order of rotational symmetry?

- 38 How many axes of symmetry does the capital letter F have?

from 6.03

- 39 Give the number of axes of symmetry and the order of rotational symmetry of the shape.



- 40 Name a quadrilateral with rotational symmetry of order 2 but no axes of symmetry.

NAME _____ CLASS _____ DATE _____

WE DO

Watch your teacher.
Copy every step.

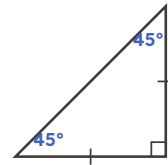
KEY IDEA

By sides: equilateral (3 equal), isosceles (2 equal), scalene (none equal). By angles: acute-angled (all less than 90°), right-angled (one 90°), obtuse-angled (one more than 90°).

E1 Classify a triangle with sides 5 cm, 5 cm and 7 cm.

E2 Classify a triangle with angles 30°, 50° and 100°.

E3 Classify the triangle by its sides and its angles.



Easy

Questions 1–9

1 Classify a triangle with angles 65°, 75° and 40° by its angles.

2 Classify a triangle with sides 8 cm, 9 cm, 14 cm by its sides.

3 How many right angles can a triangle have?

4 Classify a triangle with angles 90°, 35° and 55° by its angles.

5 Classify a triangle with sides 6 cm, 6 cm, 4 cm by its sides.

6 What size is each angle of an equilateral triangle?

7 Classify a triangle with angles 75°, 80° and 25° by its angles.

8 Classify a triangle with sides 5 cm, 5 cm, 5 cm by its sides.

9 What is the order of rotational symmetry of the capital letter F?

from 6.04

Easy

Questions 10–16

10 Classify a triangle with sides 8 cm, 10 cm, 13 cm by its sides.

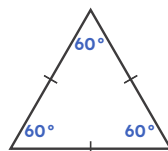
11 How many equal sides does an isosceles triangle have?

12 Classify a triangle with angles 35°, 20° and 125° by its angles.

13 $6\frac{4}{5} - 2\frac{2}{5}$

from 4.04

14 Classify the triangle by its sides and by its angles.



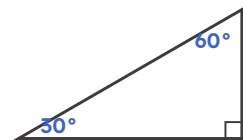
15 Classify a triangle with sides 11 cm, 13 cm, 16 cm by its sides.

16 How many equal sides does an equilateral triangle have?

Medium

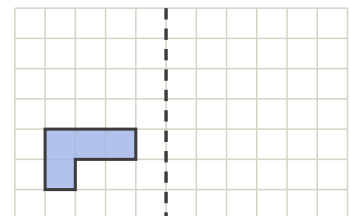
Questions 17–20

17 Classify the triangle by its sides and by its angles.

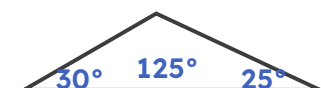


18 A triangle has angles 90°, 45° and 45°. Classify it by its angles and by its sides.

19 Reflect the shape in the dashed line. Draw the image. from 6.01



20 Classify the triangle by its sides and by its angles.



Medium

Questions 21–27

- 21 A triangle has angles 80° , 75° and 25° . Classify it by its angles and by its sides.

- 22 Classify the triangle by its sides and by its angles.

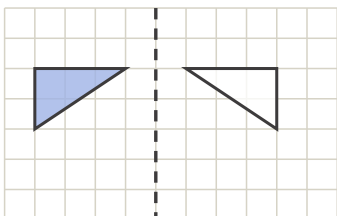


- 23 $9a \times 6x$

from 5.08

- 24 Can you draw a right-angled scalene triangle? Explain.

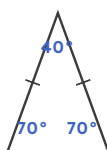
- 25 Name the transformation from the shaded shape to the white shape.



from 6.01

- 26 A triangle has angles 70° , 80° and 30° . Classify it by its angles and by its sides.

- 27 Classify the triangle by its sides and by its angles.



Challenging

Questions 28–34

- 28 Can you draw an obtuse-angled isosceles triangle? Explain.

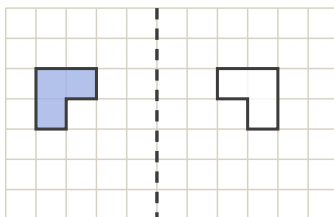
- 29 Can a triangle have sides 8 cm, 8 cm and 3 cm? Explain.

- 30 $P = 2l + 2w$. Find P when $l = 6$ and $w = 3$.

from 5.05

- 31 An isosceles triangle has two sides of 7 cm and a perimeter of 20 cm. How long is the third side?

- 32 Name the transformation from the shaded shape to the white shape.



from 6.02

- 33 Can you draw a right-angled isosceles triangle? Explain.

- 34 Can a triangle have sides 7 cm, 4 cm and 8 cm? Explain.

Challenging

Questions 35–42

- 35 Can a triangle have sides 8 cm, 4 cm and 4 cm? Explain.

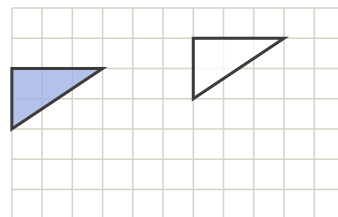
- 36 An isosceles triangle has two sides of 8 cm and a perimeter of 23 cm. How long is the third side?

- 37 Which is larger: $\frac{1}{11}$ or $\frac{1}{7}$?

from 4.02

- 38 Can you draw a triangle with two right angles? Explain.

- 39 Describe the translation from the shaded shape to the white shape.



from 6.01

- 40 A triangle has angles 80° , 35° and 65° . Classify it by its angles and by its sides.

- 41 Can a triangle have sides 9 cm, 5 cm and 3 cm? Explain.

- 42 How many axes of symmetry does the shape have?



from 6.03

NAME _____ CLASS _____ DATE _____

WE DO

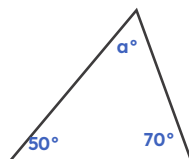
Watch your teacher.
Copy every step.

KEY IDEA

The three angles of a triangle add to 180° . To find a missing angle, take the other two from 180° . In an isosceles triangle, the two base angles are equal.

- E1** Two angles of a triangle are 65° and 80° . Find the third.

- E2** Find a .



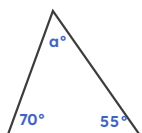
- E3** An isosceles triangle has an apex angle of 40° . Find each base angle.

Easy

Questions 1–7

- 1** Two angles of a triangle are 94° and 30° . Find the third angle.

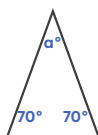
- 2** Find a .



- 3** A right-angled triangle has an angle of 57° . Find the third angle.

- 4** Two angles of a triangle are 30° and 46° . Find the third angle.

- 5** Find a .



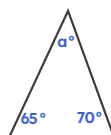
- 6** A right-angled triangle has an angle of 33° . Find the third angle.

- 7** Two angles of a triangle are 95° and 15° . Find the third angle.

Easy

Questions 8–12

- 8** Find a .



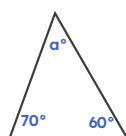
- 9** A right-angled triangle has an angle of 25° . Find the third angle.

- 10** Two angles of a triangle are 70° and 16° . Find the third angle.

- 11** How many $\frac{1}{2}$ s are in 5? Work out $5 \div \frac{1}{2}$.

from 4.07

- 12** Find a . Write the equation first.



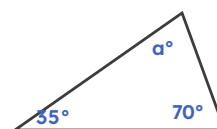
$a + ___ + ___ = ___$

$a = ___$

Medium

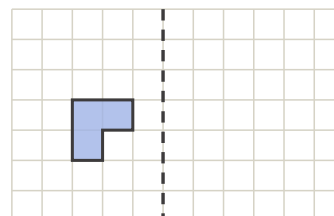
Questions 13–16

- 13** Find a . Write the equation first.



- 14** An isosceles triangle has an apex angle of 100° . Find each base angle.

- 15** Reflect the shape in the dashed line. Draw the image. from 6.01



- 16** An isosceles triangle has a base angle of 55° . Find the apex angle.

Medium

Questions 17–25

- 17 An isosceles triangle has an apex angle of 84° . Find each base angle.

- 18 An isosceles triangle has a base angle of 75° . Find the apex angle.

- 19 $\frac{3}{4}$ of 12

from 4.05

- 20 Find a . Write the equation first.



- 21 A shape is turned 90° clockwise, then 270° clockwise. What single turn does the same?

from 6.02

- 22 The angles of a triangle are x° , $3x^\circ$ and $5x^\circ$. Find x and each angle.

- 23 An isosceles triangle has an apex angle of 76° . Find each base angle.

- 24 How many axes of symmetry does the capital letter S have?

from 6.03

- 25 An isosceles triangle has a base angle of 70° . Find the apex angle.

Challenging

Questions 26–34

- 26 The angles of a triangle are $3x^\circ$, $4x^\circ$ and $5x^\circ$. Find x and each angle.

- 27 A triangle has angles x° , $(x + 40)^\circ$ and 56° . Find x .

- 28 Simplify $\frac{66}{90}$ fully.

from 4.01

- 29 Ali says a triangle has angles of 96° , 96° and 10° . Is that possible? Explain.

- 30 Angles at a point are 65° , 65° and c° . Find c .

from 2.05

- 31 The angles of a triangle are x° , x° and $2x^\circ$. Find x and each angle.

- 32 A triangle has angles x° , $(x + 10)^\circ$ and 88° . Find x .

- 33 A point is 3 squares to the left of a mirror line. Where is its image?

from 6.01

- 34 Lena says a triangle has angles of 95° , 105° and 10° . Is that possible? Explain.

Challenging

Questions 35–42

- 35 A triangle has angles x° , $(x + 10)^\circ$ and 106° . Find x .

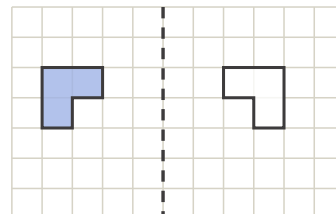
- 36 Noah says a triangle has angles of 92° , 91° and 10° . Is that possible? Explain.

- 37 $\frac{2}{4} - \frac{1}{4}$

from 4.03

- 38 The angles of a triangle are x° , $2x^\circ$ and $3x^\circ$. Find x and each angle.

- 39 Name the transformation from the shaded shape to the white shape.



from 6.02

- 40 An isosceles triangle has a base angle of 45° . Find the apex angle.

- 41 A triangle has angles x° , $(x + 30)^\circ$ and 108° . Find x .

- 42 How many axes of symmetry does a regular polygon with 12 sides have?

from 6.03

NAME _____ CLASS _____ DATE _____

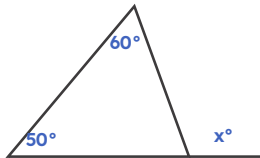
WE DO

Watch your teacher.
Copy every step.

KEY IDEA

An exterior angle of a triangle is made by extending one side. The exterior angle equals the sum of the two interior opposite angles.

- E1** Find x , the exterior angle.



- E2** The interior opposite angles are 35° and 75° . Find the exterior angle.

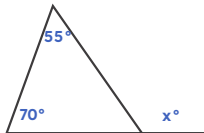
- E3** The exterior angle is 130° . One interior opposite angle is 45° . Find the other.

Easy

Questions 1–7

- 1** The interior opposite angles are 82° and 62° . Find the exterior angle.

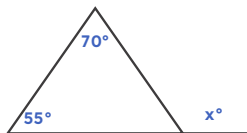
- 2** Find x , the exterior angle.



- 3** An interior angle of a triangle is 142° . Find the exterior angle next to it.

- 4** The interior opposite angles are 59° and 20° . Find the exterior angle.

- 5** Find x , the exterior angle.



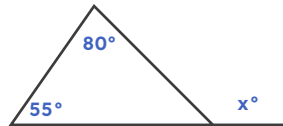
- 6** An interior angle of a triangle is 115° . Find the exterior angle next to it.

- 7** The interior opposite angles are 51° and 85° . Find the exterior angle.

Easy

Questions 8–13

- 8** Find x , the exterior angle.



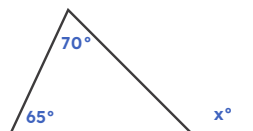
- 9** An interior angle of a triangle is 73° . Find the exterior angle next to it.

- 10** The interior opposite angles are 67° and 61° . Find the exterior angle.

11 $\frac{10}{11} - \frac{1}{11}$

from 4.03

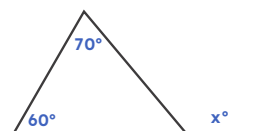
- 12** Find x . Give the rule.



$x = ___ + ___$

$x = ___ \text{ (exterior angle)}$

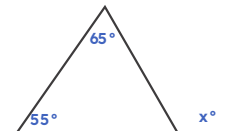
- 13** Find x , the exterior angle.



Medium

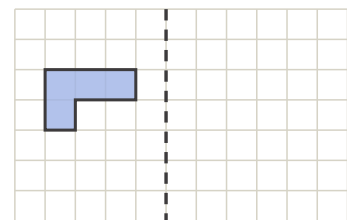
Questions 14–17

- 14** Find x . Give the rule.

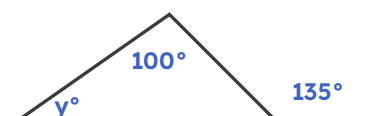


- 15** The exterior angle is 110° and one interior opposite angle is 35° . Find the other.

- 16** Reflect the shape in the dashed line. from 6.01
Draw the image.



- 17** Find y , then check with the angle sum.

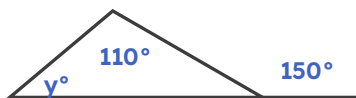


Medium

Questions 18–25

- 18 The exterior angle is 75° and one interior opposite angle is 65° . Find the other.

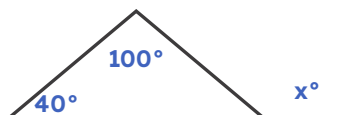
- 19 Find y , then check with the angle sum.



- 20 $\frac{2}{3}$ of 24

from 4.05

- 21 Find x . Give the rule.



- 22 A shape is turned 90° clockwise, then 270° clockwise. What single turn does the same?

from 6.02

- 23 The interior opposite angles are x° and $(x + 15)^\circ$. The exterior angle is 81° . Find x .

- 24 The exterior angle is 160° and one interior opposite angle is 75° . Find the other.

- 25 A right-angled triangle has an angle of 19° . Find the third angle.

from 6.06

Challenging

Questions 26–33

- 26 The interior opposite angles are x° and $(x + 5)^\circ$. The exterior angle is 43° . Find x .

- 27 An isosceles triangle has base angles of 55° . Find the exterior angle at a base vertex.

- 28 Write $8km$ in expanded form.

from 5.03

- 29 What do the three exterior angles of a triangle (one at each corner) add to?

- 30 What is the order of rotational symmetry of a regular polygon with 10 sides?

from 6.04

- 31 The interior opposite angles are x° and $(x + 15)^\circ$. The exterior angle is 55° . Find x .

- 32 An isosceles triangle has base angles of 68° . Find the exterior angle at a base vertex.

- 33 How many axes of symmetry does the word DAD have (in capitals)?

from 6.03

Challenging

Questions 34–40

- 34 An isosceles triangle has base angles of 42° . Find the exterior angle at a base vertex.

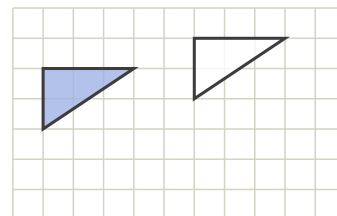
- 35 Can an exterior angle of a triangle be acute? Explain.

- 36 $kp \div k$

from 5.09

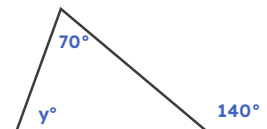
- 37 The interior opposite angles are x° and $(x + 20)^\circ$. The exterior angle is 80° . Find x .

- 38 Name the transformation from the shaded shape to the white shape.



from 6.01

- 39 Find y , then check with the angle sum.



- 40 An isosceles triangle has base angles of 51° . Find the exterior angle at a base vertex.

NAME _____ CLASS _____ DATE _____

WE DO

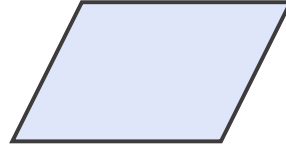
Watch your teacher.
Copy every step.

KEY IDEA

Quadrilaterals have four sides. Square, rectangle, rhombus and parallelogram all have two pairs of parallel sides. A trapezium has exactly one pair. A kite has two pairs of equal sides next to each other.

E1 Name the quadrilateral: 4 equal sides, no right angles.

E2 Name the quadrilateral and give one property.



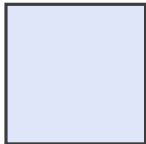
E3 True or false: every square is a rectangle.

Easy

Questions 1–7

1 Name the quadrilateral: 4 equal sides and 4 right angles.

2 Name the quadrilateral.



3 How many pairs of parallel sides does a trapezium have?

4 Name the quadrilateral: 4 right angles, and opposite sides equal.

5 Name the quadrilateral.



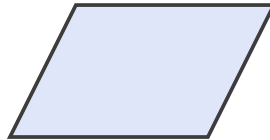
6 How many right angles does a rectangle have?

7 Name the quadrilateral: 4 equal sides and no right angles.

Easy

Questions 8–13

8 Name the quadrilateral.



9 How many sides does a quadrilateral have?

10 Name the quadrilateral: 2 pairs of parallel sides and no right angles.

11 $8b \times 9y$

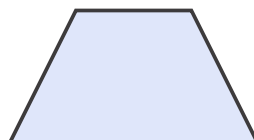
from 5.08

12 Name the quadrilateral, then give one property.



all sides equal and all angles ____°

13 Name the quadrilateral.



Medium

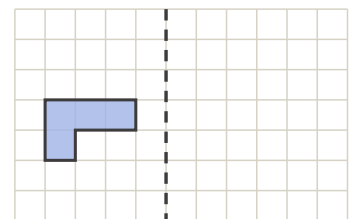
Questions 14–17

14 Name the quadrilateral, then give one property.

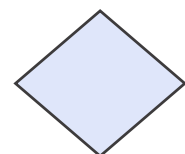


15 True or false? Every square is a rectangle.

16 Reflect the shape in the dashed line. Draw the image.



17 Name the quadrilateral, then give one property.

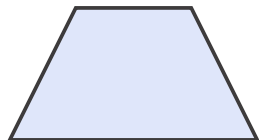


Medium

Questions 18–26

- 18 True or false? A kite is a parallelogram.

- 19 Name the quadrilateral, then give one property.



- 20 Which is larger: $\frac{1}{12}$ or $\frac{1}{7}$?

from 4.02

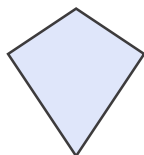
- 21 My opposite sides are parallel and I have a right angle. What two shapes could I be?

- 22 A shape is turned 90° clockwise, then 180° clockwise. What single turn does the same?

from 6.02

- 23 True or false? Every square is a rhombus.

- 24 Name the quadrilateral, then give one property.



- 25 How many equal sides does an isosceles triangle have?

from 6.05

- 26 I have two pairs of equal sides next to each other and no parallel sides. What am I?

Challenging

Questions 27–36

- 27 I have one pair of parallel sides and two equal sides that are not parallel. What am I?

- 28 Explain why a square is also a rectangle.

- 29 4 equal packs hold 96 pencils. How many pencils are in each pack?

from 5.13

- 30 Which quadrilaterals have both pairs of opposite sides parallel?

- 31 A right-angled triangle has an angle of 73° . Find the third angle.

from 6.06

- 32 I have 4 equal sides. My diagonals are equal. What am I?

- 33 Explain why a rhombus is also a parallelogram.

- 34 How many axes of symmetry does the word DAD have (in capitals)?

from 6.03

- 35 Which quadrilaterals have all four sides equal?

- 36 A triangle has angles 50° , 45° and 85° . Classify it by its angles and by its sides.

from 6.05

Challenging

Questions 37–39

- 37 Explain why a square is also a rhombus.

- 38 Which quadrilaterals have four right angles?

- 39 $3 \times \frac{3}{8}$

from 4.06

6.09 Angle sum of a quadrilateral

YEAR 7 · CHAPTER 6: GEOMETRICAL FIGURES · PAGE 1 OF 2

NAME _____ CLASS _____ DATE _____

WE DO

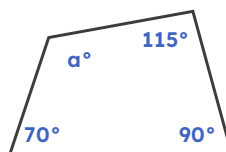
Watch your teacher.
Copy every step.

KEY IDEA

The four angles of a quadrilateral add to 360° : it can be cut into two triangles. To find a missing angle, take the other three from 360° .

- E1** Three angles of a quadrilateral are 80° , 95° and 110° . Find the fourth.

- E2** Find a .



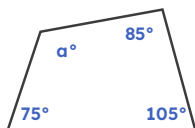
- E3** A parallelogram has an angle of 65° . Find the other three angles.

Easy

Questions 1–7

- 1** Three angles of a quadrilateral are 135° , 55° and 140° . Find the fourth.

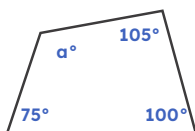
- 2** Find a . (Not to scale.)



- 3** What size is each angle of a rectangle?

- 4** Three angles of a quadrilateral are 55° , 70° and 130° . Find the fourth.

- 5** Find a . (Not to scale.)



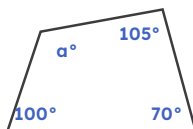
- 6** What do the angles of a triangle add to?

- 7** Three angles of a quadrilateral are 70° , 125° and 60° . Find the fourth.

Easy

Questions 8–13

- 8** Find a . (Not to scale.)



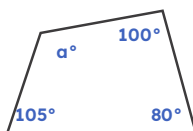
- 9** What do the angles of a quadrilateral add to?

- 10** Three angles of a quadrilateral are 75° , 95° and 125° . Find the fourth.

- 11** Write 5 out of 12 as a fraction in simplest form.

from 4.12

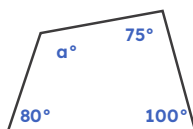
- 12** Find a . Write the equation first.



$$a + ___ = ___$$

$$a = ___$$

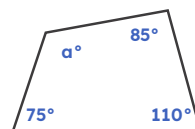
- 13** Find a . (Not to scale.)



Medium

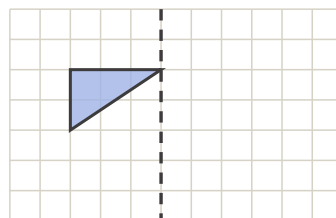
Questions 14–17

- 14** Find a . Write the equation first.



- 15** A parallelogram has an angle of 45° . Find the other three angles.

- 16** Reflect the shape in the dashed line. Draw the image.



- 17** A kite has two equal angles x° . Its other angles are 115° and 55° . Find x .

Medium

Questions 18–25

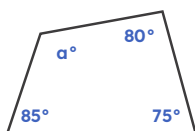
- 18 A parallelogram has an angle of 145° . Find the other three angles.

- 19 A kite has two equal angles x° . Its other angles are 70° and 100° . Find x .

- 20 Write an expression: 4 less than k

from 5.04

- 21 Find a . Write the equation first.



- 22 A shape is turned 180° clockwise, then 180° clockwise. What single turn does the same?

from 6.02

- 23 The angles of a quadrilateral are in the ratio $3 : 4 : 5 : 6$. Write them as $3x^\circ$, $4x^\circ$, $5x^\circ$, $6x^\circ$ and find x .

- 24 A parallelogram has an angle of 50° . Find the other three angles.

- 25 A right-angled triangle has an angle of 70° . Find the third angle.

from 6.06

Challenging

Questions 26–34

- 26 The angles of a quadrilateral are in the ratio $1 : 2 : 3 : 4$. Write them as x° , $2x^\circ$, $3x^\circ$, $4x^\circ$ and find x .

- 27 A quadrilateral has angles x° , $2x^\circ$, $(x + 30)^\circ$ and 166° . Find x .

- 28 Use a calculator: $6\frac{2}{9} - 1\frac{5}{7}$

from 4.09

- 29 Can a quadrilateral have three obtuse angles? Explain.

- 30 An interior angle of a triangle is 61° . Find the exterior angle next to it.

from 6.07

- 31 The angles of a quadrilateral are in the ratio $2 : 3 : 3 : 4$. Write them as $2x^\circ$, $3x^\circ$, $3x^\circ$, $4x^\circ$ and find x .

- 32 A quadrilateral has angles x° , $2x^\circ$, $(x + 10)^\circ$ and 162° . Find x .

- 33 How many axes of symmetry does the word WOW have (in capitals)?

from 6.03

- 34 Explain why the angles of a quadrilateral add to 360° .

Challenging

Questions 35–43

- 35 A quadrilateral has angles x° , $2x^\circ$, $(x + 40)^\circ$ and 164° . Find x .

- 36 Can a quadrilateral have four obtuse angles? Explain.

- 37 $\frac{6}{8} - \frac{2}{8}$

from 4.03

- 38 The angles of a quadrilateral are in the ratio $1 : 2 : 2 : 3$. Write them as x° , $2x^\circ$, $2x^\circ$, $3x^\circ$ and find x .

- 39 What is the order of rotational symmetry of the capital letter B?

from 6.04

- 40 A kite has two equal angles x° . Its other angles are 90° and 60° . Find x .

- 41 A quadrilateral has angles x° , $2x^\circ$, $(x + 10)^\circ$ and 170° . Find x .

- 42 Name the quadrilateral: 4 equal sides and 4 right angles.

from 6.08

- 43 The angles of a quadrilateral are in the ratio $1 : 1 : 2 : 2$. Write them as x° , x° , $2x^\circ$, $2x^\circ$ and find x .

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

Learn the properties: sides (equal? parallel?), angles (equal? right?) and diagonals (equal? meet at 90° ? bisect each other?). Use them to find missing sides and angles.

Watch your teacher.
Copy every step.

E1 A rectangle has one diagonal 13 cm long.
How long is the other?

E2 A rhombus has a side of 7 cm. Find its perimeter.

E3 Do the diagonals of a kite meet at right angles?

Easy

Questions 1–9

1 True or false? The diagonals of a rectangle are equal.

2 A rhombus has a side of 11 cm. Find its perimeter.

3 A rectangle has one diagonal 4 cm long. How long is the other?

4 True or false? The diagonals of a rhombus meet at 90° .

5 A rhombus has a side of 5 cm. Find its perimeter.

6 A rectangle has one diagonal 6 cm long. How long is the other?

7 True or false? The diagonals of a parallelogram are equal.

8 A rhombus has a side of 7 cm. Find its perimeter.

9 Name the quadrilateral.



from 6.08

Easy

Questions 10–17

10 A rhombus has a side of 3 cm. Find its perimeter.

11 A rectangle has one diagonal 15 cm long. How long is the other?

12 True or false? A square has equal diagonals that meet at 90° .

13 $\frac{2}{3}$ of 6

from 4.05

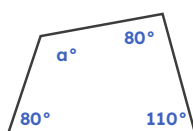
14 A square has a perimeter of 20 cm. Find a side.

_____ ÷ _____ = _____ cm

15 A rhombus has a side of 4 cm. Find its perimeter.

16 A rectangle has one diagonal 14 cm long. How long is the other?

17 Find a . (Not to scale.)



from 6.09

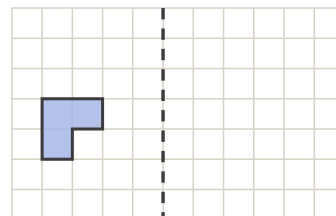
Medium

Questions 18–23

18 A rhombus has an angle of 55° . Find the opposite angle.

19 Describe the diagonals of a parallelogram.

20 Reflect the shape in the dashed line. Draw the image. from 6.01



21 A rectangle is 10 cm long and 13 cm wide. Find its perimeter.

22 Describe the diagonals of a kite.

23 The exterior angle is 70° and one interior opposite angle is 15° . Find the other.

from 6.07

Medium

Questions 24–33

- 24 Describe the diagonals of a rhombus.
-
-
- 25 A square has a perimeter of 48 cm. Find a side.
-
-
- 26 Write 29% as a decimal.
-
- from 4.10
- 27 A quadrilateral has diagonals that bisect each other but are not equal and do not meet at 90° . What is it?
-
- 28 A shape is turned 180° clockwise, then 180° clockwise. What single turn does the same?
-
- from 6.02
- 29 Describe the diagonals of a square.
-
-
- 30 A rhombus has an angle of 80° . Find the opposite angle.
-
-
- 31 How many pairs of parallel sides does a trapezium have?
-
- from 6.08
- 32 A quadrilateral has equal diagonals that bisect each other but do not meet at 90° . What is it?
-
-
- 33 A kite has two equal angles x° . Its other angles are 95° and 45° . Find x .
-
- from 6.09

Challenging

Questions 34–42

- 34 A quadrilateral has equal diagonals that meet at 90° and bisect each other. What is it?
-
-
- 35 The diagonals of a rhombus are 12 cm and 16 cm. How far is it from the centre to each corner?
-
-
- 36 $24 + 41 + 76 + 85$
-
- from 5.01
- 37 Give a reason: A square is a rectangle.
-
-
- 38 An interior angle of a triangle is 59° . Find the exterior angle next to it.
-
- from 6.07
- 39 The diagonals of a rhombus are 8 cm and 6 cm. How far is it from the centre to each corner?
-
-
- 40 Give a reason: A square is a rhombus.
-
-
- 41 How many axes of symmetry does the word BOB have (in capitals)?
-
- from 6.03
- 42 The diagonals of a rhombus are 10 cm and 12 cm. How far is it from the centre to each corner?
-
-

Challenging

Questions 43–51

- 43 The diagonals of a rhombus are 10 cm and 6 cm. How far is it from the centre to each corner?
-
-
- 44 Give a reason: A rhombus is a parallelogram.
-
-
- 45 Write an expression: 9 less than a
-
- from 5.04
- 46 The diagonals of a rhombus are 12 cm and 14 cm. How far is it from the centre to each corner?
-
-
- 47 What is the order of rotational symmetry of the capital letter P?
-
- from 6.04
- 48 Give a reason: A rhombus is a kite.
-
-
- 49 The diagonals of a rhombus are 12 cm and 6 cm. How far is it from the centre to each corner?
-
-
- 50 Three angles of a quadrilateral are 60° , 110° and 100° . Find the fourth.
-
- from 6.09
- 51 The diagonals of a rhombus are 14 cm and 10 cm. How far is it from the centre to each corner?
-
-